
CHAPTER 1

VECTOR SPACES AND LINEAR EQUATIONS

1 Basic Structures and Definitions

Definition 1.1: Ring R

A set R with two operations $+: R \times R \rightarrow R$ and $\cdot: R \times R \rightarrow R$ is called a ring if the following hold:

1. (Additive abelian group) $(R, +)$ is an abelian group with identity 0.
2. (Distributivity) $x(a + b) = x \cdot a + x \cdot b$ and $(x + y)a = x \cdot a + y \cdot a$

Definition 1.2: Field

$(\mathbb{F}, +, \cdot)$ is a field if $\mathbb{F}$ is a set with operations $+: \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$ and $\cdot: \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$ such that the following hold:

1. $(\mathbb{F}, +)$ is an abelian group with an identity element 0, i.e. $\forall a, b, c \in \mathbb{F}$, we have $a + (b + c) = (a + b) + c$ (Associativity), $a + b = b + a$ (Commutativity), $\exists 0 \in \mathbb{F}$ such that $a + 0 = a$, and $\forall a \in \mathbb{F}$, $\exists -a \in \mathbb{F}$ such that $a + (-a) = 0$.
2. $(\mathbb{F} \setminus \{0\}, \cdot)$ is also an abelian group, i.e. $\forall a, b, c \in \mathbb{F} \setminus \{0\}$, we have $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (Associativity), $a \cdot b = b \cdot a$, $\exists 1$ such that $a \cdot 1 = a$ and $\forall a \in \mathbb{F} \setminus \{0\}$ there exists a unique a^{-1} such that $aa^{-1} = 1$.
3. (Distributivity) $\forall a, b, c \in \mathbb{F}$, $a(b + c) = ab + ac$.

Such a structure is called a *Field*.

1.1 Linear Equations and Matrices

Definition 1.3: System of Linear Equation

We call a system of linear equations a collection of equations of the form:

$$a_{11}x_1 + a_{12}x_2 \cdots + a_{1n}x_n = y_1$$

$$a_{21}x_1 + a_{22}x_2 \cdots + a_{2n}x_n = y_2$$

$$\vdots$$

$$a_{i1}x_1 + a_{i2}x_2 \cdots + a_{in}x_n = y_i$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 \cdots + a_{mn}x_n = y_m$$

where $a_{ij}, y_j \in \mathbb{F}$. This system has n variables x_j taking values (if at all possible) in $\mathbb{F}$ and m equations.

In convenient matrix notation this reads:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$